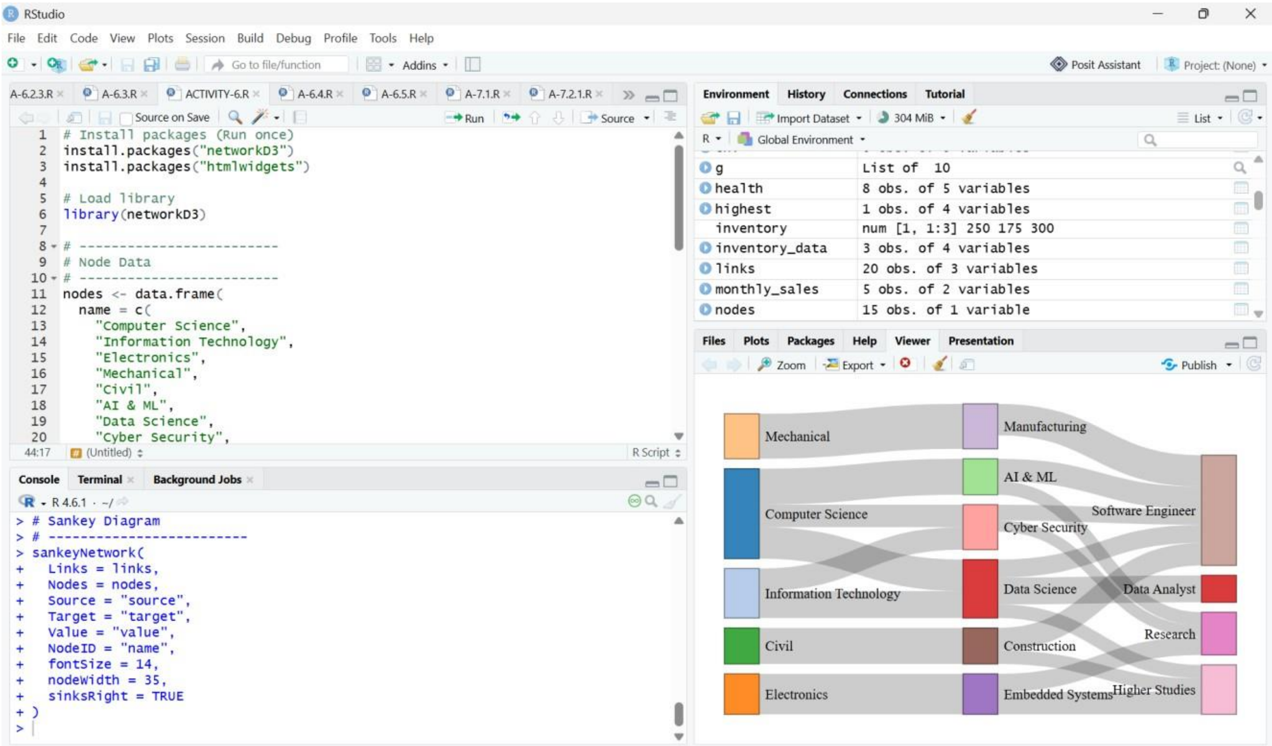
ACTIVITY-6

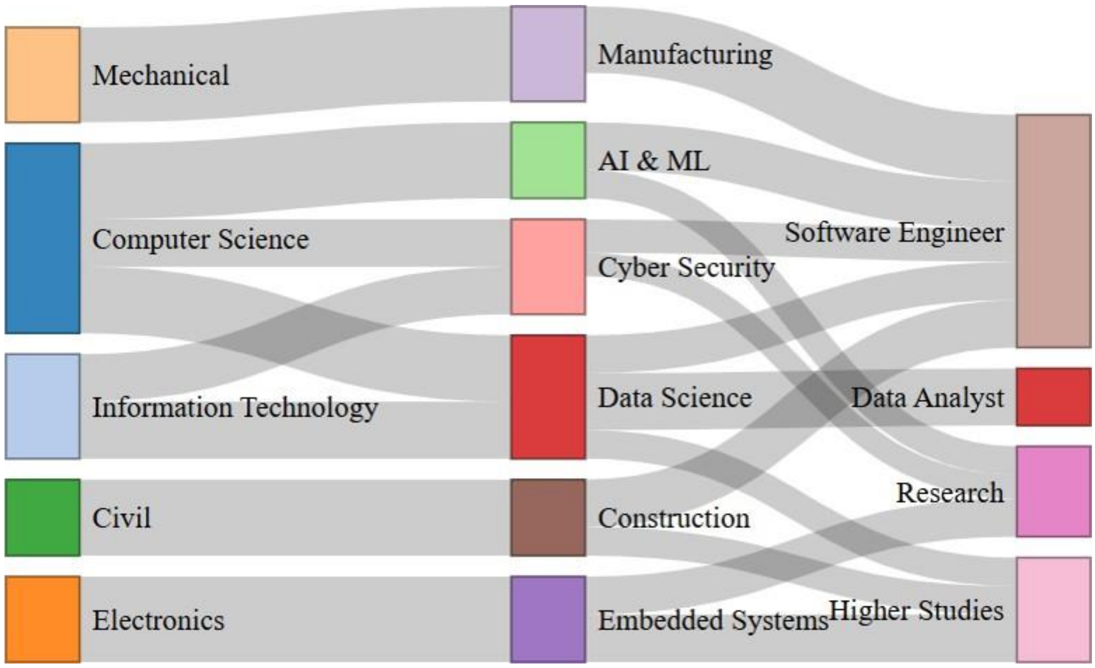


Code:

```
# Install packages (Run once)

install.packages("networkD3")

install.packages("htmlwidgets")

# Load library

library(networkD3)

# Node Data

nodes <- data.frame(

  name = c(

    "Computer Science",

    "Information Technology",

    "Electronics",

    "Mechanical",

    "Civil",
```

```
    "AI & ML",
    "Data Science",
    "Cyber Security",
    "Embedded Systems",
    "Manufacturing",
    "Construction",
    "Software Engineer",
    "Data Analyst",
    "Research",
    "Higher Studies"
  )
)
# Link Data
links <- data.frame(
  source = c(0,0,0,1,1,2,3,4,5,5,6,6,6,7,7,8,8,9,10,10),
  target = c(5,6,7,6,7,8,9,10,11,13,11,12,14,11,13,13,14,11,11,14),
  value  = c(80,70,50,60,50,90,100,80,50,30,40,60,30,35,25,40,50,70,50,30)
)
# Sankey Diagram
sankeyNetwork(
Links = links,
Nodes = nodes,
Source = "source",
Target = "target",
Value = "value",
NodeID = "name",
fontSize = 14,
  nodeWidth = 35,
  sinksRight = TRUE)
```

Output:



2. Configuration Used

Parameter	Value
Node Width	**35**
Font Size	**14**
Career Outcomes Position	**Right Side (sinksRight = TRUE)**

3. Interpretation

(a) Specialization Receiving the Highest Number of Students

Calculate total inflow into each specialization.

Specialization	Students
AI & ML	80
Data Science	**130**
Cyber Security	100
Embedded Systems	90

Specialization	Students
Manufacturing	100
Construction	80

Answer: Data Science (130 students)

(b) Career Outcome with Largest Overall Inflow

Career Outcome	Total Students
Software Engineer	**245**
Data Analyst	60
Research	95
Higher Studies	110

Answer: Software Engineer (245 students)

(c) Department Contributing the Most Students to Software Engineer

Tracing Software Engineer flow:

- Computer Science
  - AI & ML → Software Engineer = 50
  - Data Science → Software Engineer = 40
  - Cyber Security → Software Engineer = 35
  - **Total = 125**
- Information Technology
  - Data Science contributes part of 40
  - Cyber Security contributes part of 35
  - (Less than Computer Science)
- Mechanical
  - Manufacturing → Software Engineer = 70
- Civil
  - Construction → Software Engineer = 50

- Electronics
    - No direct Software Engineer flow

Answer: Computer Science contributes the most students to Software Engineer positions.

4. Why Sankey Diagram is Better than Bar Chart or Pie Chart

Sankey Diagram Advantages

- Clearly shows the **flow of students** from Departments → Specializations → Career Outcomes.
- Displays relationships between multiple stages in one visualization.
- Link thickness represents the number of students.
- Makes it easy to trace career pathways from start to finish.
- Highlights major and minor transitions visually.

Why Not a Bar Chart?

- Shows only numerical comparisons.
- Cannot represent movement between multiple categories.
- Does not show connections between departments, specializations, and careers.

Why Not a Pie Chart?

- Displays proportions only.
- Cannot illustrate hierarchical flow.
- Difficult to compare multiple stages simultaneously.

5. Two Improvements for University Administrators

Improvement 1

Use color coding for each department or specialization to make pathways easier to distinguish.

Improvement 2

Add interactive tooltips displaying:

- Department name
- Specialization

- Career outcome
- Exact number of students
- Percentage of total students

This allows administrators to analyze pathways more effectively.

Expected Output Interpretation

- The Sankey diagram illustrates the complete flow of students from **Departments → Specializations → Career Outcomes**.
- **Data Science** receives the highest number of students (**130**).
- **Software Engineer** is the most common career outcome (**245 students**).
- **Computer Science** contributes the largest number of students to **Software Engineer** roles.
- The visualization helps identify dominant academic pathways and supports better curriculum planning and career guidance.